

# GOKULANATH K

Software Developer | Computer Vision & Assistive Tech Specialist

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## PROFESSIONAL SUMMARY

Resourceful Software Developer and MCA Candidate specializing in Computer Vision, Human-Computer Interaction (HCI), and Assistive Technology. Hands-on experience building AI rehabilitation platforms, real-time facial recognition systems, and gesture/eye-blink accessibility tools. Published international researcher (CAIT 2026) and Smart India Hackathon award winner with core strength in Python, Flask, OpenCV, MediaPipe, Scikit-Learn, SQL, and Full-Stack Web Architecture.

## TECHNICAL SKILLS

- Languages & DBs:** Python, Java, C++, SQL (MySQL, SQLite), JavaScript (ES6+), HTML5, CSS3
- Computer Vision & AI:** OpenCV, MediaPipe, Pose Estimation, Scikit-Learn, Facial Recognition, EAR Algorithm
- Web & DevOps Tools:** Flask, React.js, Node.js, Express, RESTful APIs, Telegram Bot API, Docker, Git/GitHub
- Core Competencies:** Digital Rehabilitation Tech, Hands-Free HCI, Machine Learning, Agile Collaboration

## KEY PROJECTS

### RehabCare – AI-Powered Digital Rehabilitation Platform 2026

**Technologies:** Python, Flask, OpenCV, MediaPipe, Scikit-Learn, Telegram Bot API

- Architected an end-to-end digital rehabilitation suite scoring patient exercise technique and counting repetitions in real-time via live webcams or video uploads using MediaPipe pose estimation.
- Engineered dual hands-free communication channels (hand-gesture recognition & eye-blink Morse code) enabling speech/motor-impaired patients to interact directly with medical care teams.
- Trained machine learning models to dynamically recommend next-session exercise difficulty based on historical recovery metrics, integrated with a live Telegram bot for real-time doctor alerts.

### Customized Eye-Blink Assistive Communication System for Motor-Impaired Patients 2025 – 2026

**Technologies:** Python, OpenCV, Eye Aspect Ratio (EAR) Algorithm, Telepresence Integration

- Engineered a low-cost, hands-free assistive communication system utilizing webcam-based real-time eye-blink tracking to convert blink patterns into emergency alerts and messages.
- Integrated doctor interaction logging and telepresence alerting, significantly reducing patient response time and setup costs compared to expensive commercial alternatives.

### Real-Time Hand Gesture-Based Mouse & Computer Control 2025

**Technologies:** Python, OpenCV, MediaPipe, PyAutoGUI

- Developed a touchless computer control interface tracking 21 hand landmarks via webcam to map hand movements directly to OS cursor movement, clicks, and window controls.

### Face Recognition with Dynamic User Registration 2024 – 2025

**Technologies:** Python, OpenCV, Deep Learning Vector Embeddings, SQLite

- Built an intelligent face recognition pipeline that identifies users in live video feeds and registers new profiles on-the-fly without requiring full model retraining or system downtime.

## HONORS & ACHIEVEMENTS

- Best Idea Award:** Secured 1st Place at the Inter-College round of **Smart India Hackathon (SIH 2025)** for innovative IoT & computer vision assistive technology.
- Published International Research:** Authored paper *"Customized Eye Blink Based Communication System for Paralysed Person"* published in the **7th International Conference on Computing Analytics and Information Technology (CAIT 2026)**.

## EDUCATION

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|--------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>Master of Computer Applications (MCA)</b><br>Hindusthan College of Arts and Science, Coimbatore, TN       | 2025 – Present<br><b>CGPA: 7.9</b> |
| <b>Bachelor of Computer Applications (BCA)</b><br>Hindusthan College of Arts and Science, Coimbatore, TN     | 2022 – 2025                        |
| <b>Higher Secondary Certificate (12th Grade)</b><br>NM Matriculation Higher Secondary School                 | 2021 – 2022                        |
| <b>Secondary School Leaving Certificate (10th Grade)</b><br>Kalaimagal Matriculation Higher Secondary School | 2019 – 2020                        |